

Highlights

- Morphoelastic-rod model of root growth into a Mohr-Coulomb granular medium
- Gravity enters via self-weight, granular strength, and statolith gain
- Exact penetration law: burial depth scales inversely with gravity ($z^* \sim 1/g$)
- Numerics confirm the law over a 30-fold range of gravitational acceleration
- Gravitropism and circumnutation act as independent depth and shape controls